

Inclusive Community Accessibility and Neurodiversity in Classroom Industry and Academia

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ABSTRACT

Universal design/accessibility, inclusive classrooms and workplaces are essential components for the success of individuals affected by various challenges. Neurodivergent and disabled individuals have a variety of needs that are often unmet by schools and industries in the SIGGRAPH community. This panel discusses the current state of the issue, the challenges of neurodivergent and disabled individuals, different types of accommodations and their availability and effectiveness, and helpful ideas to make the learning and working environments more inclusive.

Although accommodations for some issues are required by law in the United States (including Americans with Disability Act (ADA) compliance), levels of enforcement and awareness vary. Moreover, accommodations to provide inclusive classrooms are not necessarily required in other jurisdictions. Yet, we know that persons who face different types of challenges are still able to make significant contributions to whatever endeavors in which they happen to be involved and there is an awareness of the importance of creating inclusive classrooms and workplaces that accommodate individuals with neurodiversity as well as people with various (and sometimes more obvious) physical disabilities. The panel discusses different types of accommodations, and what may be available for classrooms and workplaces.

Students in tertiary education are coming from a system where supervising adults (parents, guardians, counselors) worked on their behalf to ensure they received the accommodations needed in primary and secondary schools (generally K-12 in the United States). Upon arrival at university, they often do not know how to self-advocate and parents or guardians, no longer able (or allowed) to

advocate on behalf of their young adults, don't know how to support them effectively. Meanwhile, faculty, administrators, and staff may not have been trained in effective methods of accommodating the variety of students in a classroom. Challenges that all students face are sometimes more difficult for those with challenges including how to manage time, effectively study, and manage interactions with instructors and classmates. They also need to learn about school policies, procedures and resources that may be available to help them. These questions and concerns are being addressed in several different ways, in the classroom, through civic engagement, and within the workplace.

Some accommodations are for physical challenges, and others for those who are neurodivergent. These accommodations can sometimes overlap and can benefit both. For example, scaffolding assignments can help all students be less overwhelmed with a project, by breaking it into smaller milestone goals. Scaffolding projects can be mirrored in the workforce as well by allowing employees to have several small goals that lead to the final larger goal and can give room for feedback and possible iteration. Other strategies that can be employed include pre-recording lectures when intellectual property issues are not being violated. Offering Lectures as PDFs in the Learning Management System your class uses allows for students to have access to the relevant notes of the in-class lectures and accommodates any students who require lecture notes to help them succeed. Setting up auto-reminders to the class via email regarding assignments and project deadlines can help all students with their time-management skills and meeting project deadlines.

Neurodivergent students [Asbell-Clarke et al. 2022], however, have a range of different challenges that need to be accommodated and may include issues of noise, light, learning style(s), sensitivity to issues of personal space, and ability to process information.

It has been announced that providing a place with a relaxed environment is effective for people with mental disorders such as autism and developmental disorders. For example, at SIGGRAPH 23, Inclusive Quiet Room was exhibited as a product for people with mental disabilities [Kimura et al. 2023]. This preaches that

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even if there are people with mental disorders in the SIGGRAPH community. It as a community should be tolerant of diversity and accept everyone. This product can also be applied to schools.

Even a so-called single issue, like dyslexia, may range from mild to severe. Topics such as active learning, breaking apart and/or flexibility in assignments and projects, physical/spatial accommodations, attendance policies, industry-academy partnerships, and civic engagement are all topics of discussion.

Establishing a welcoming atmosphere for all forms of diversity in the classroom and the studio is essential for student and employee success. This includes recognizing and addressing additional needs of individuals like those with PTSD, veterans, current military members, and parents of young children. Creating a supportive environment means understanding and accommodating their unique challenges, such as providing hybrid attendance policies, seating near the exit doors, regular breaks, clear safety procedures, and listing availability for mental health and childcare resources. Examples of ways to help some of those with these types of issues are discussed by the panel.

Schools and companies may believe that they do not have the wherewithal to effectively accommodate a specialized population, leading to questions such as, what is the reality of working in an industry known for being deadline-driven and collaborative? How can we foster more contributions from neurodiverse individuals to the community? Excerpts from short interviews with industry professionals are included as part of the panel.

Supporting neurodivergent and disabled students and employees requires a comprehensive approach [Junko Saimoto 2023], [John Hopkins University 2022] that can benefit all. By implementing these strategies, we create more inclusive and effective educational and professional environments. However, it is essential to continuously assess and improve practices, staying informed about new research and emerging technologies that can further aid inclusivity. Future efforts could include developing more advanced assistive technologies [Boyd 2023], offering specialized training programs for faculty and staff, and fostering partnerships with organizations specializing in neurodiversity and disability services. It is an explicit goal of this panel to create/improve a culture of inclusivity by promoting ongoing dialogue within the SIGGRAPH community, encouraging feedback, and adapting to the evolving needs of our evolving and increasingly diverse population. By prioritizing inclusivity, we not only support the success of individuals with unique challenges but also enrich our community as a whole, fostering innovation and creativity through diverse perspectives and experiences.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**; *Accessibility theory, concepts and paradigms.*

KEYWORDS

Education, Inclusive Classroom, Disabilities, Neurodiversity

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